**Quantum School of Technology**

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# Syllabus

**Course Type:** Computer Applications

**Course :** **Bachelor of Computer Applications**

**Year/sem:** **6 Sem**

**Branch :**

**Created By:**

**Fundamentals of Computer Organization**

**CA36301**

| Unit | Sub Unit | Topic | Duration |
|---|---|---|---|
| **Unit I** | | Functional units of digital systems & their interconnections, Register Transfer Language,, Bus andMemoryTransfers, Bus Architecture,, Bus Arbitration, Arithmetic Logic, Shift Micro operation, ArithmeticLogicShift Unit, Booth Multiplication Algorithm, IEEE standard for Floating point numbers. | 8 |
| **Unit II** | | Instruction types, formats, Instruction cycles & sub cycles, Fundamental Concepts (Register Transfers,, performing of arithmetic or logical operations, fetching a word from memory, Storingaword in memory Hardwired Control, Micro programmed control | 7 |
| **Unit III** | | Processor Organization: General register organization, Stack organization, Addressing mode, Datatransfer&Manipulations, Pipelining-Introduction, linear pipe-line processor | 7 |
| **Unit IV** | | Peripheral devices, Input-Output Interface, I/O ports, Modes Of Transfer, Types of interrupts &exceptions,, Priority Interrupt, DMA, Input-Output Processor (IOP) | 7 |
| **Unit V** | | Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory | 7 |

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